

Presentation Supplements for Greens Function School

Rohit Goswami · 2021-09-10 · presentations · ramblings · fortran

A meta-post on an oral presentation around Fortran and languages

Background

I was a participant at the 2021 Les Houches school of physics on “Green’s function approach to multiple scattering theory in electronic structure and spectroscopies”. I opted to give a student talk on programming languages and elementary functions as a cautionary tale for IEEE 754.

Details

Title

Programming Language Interstices

Abstract

In this short talk I will discuss the changing landscape of programming languages from the viewpoint of “mixed language” bindings. That is, the representations of various programming languages and their standardisations to aid making library calls and generating wrappers between languages. In particular I will demonstrate the LFortran compilers’ ASR (Abstract Semantic Representation) and outline its usage for generating strictly correct Python wrappers; and discuss recent changes and future plans for the venerable f2py wrapper generator. f2py is currently part of Python’s numpy and is best known for being used extensively to generate scipy’s calls to BLAS and LAPACK; thus it forms a core of the fast growing Scientific Python ecosystem. Error analysis and the pitfalls of precision take a central role. Time permitting, some discussion of the Python-C and Fortran-C bindings will also take place. The context of these is to be concrete use cases directly applicable to the applied computational sciences.

Other Content

Anything tagged Fortran on this site.

Slides

- [Best viewed here](#) using a browser (in a new tab)
- A `pdf` copy of the slides are embedded below
- The `orgmode` source [is here on the site’s GH repo](#)

Video

TBD

Thoughts

This was the first time in a long while since I gave an in-person short presentation. I should have used a timer.